

8. The Address LEDs now show the address of the ROM being tested and the Data LEDs show the computed checksum. Compare this checksum against the list of correct checksums for the version of 4051 firmware being tested. These checksums are found in Table 5-3.
9. Press START to continue the test to the next ROM address. Cycle through steps 8 and 9 until all ROM devices have been checked.

Table 5-3

4051 FIRMWARE CHECKSUMS

Part Number	Circuit Number	Starting Address	Version 1	Version 2	Version 3	Version 4	Version 5	Version 6	Version 6	
									Circuit Number	Part Number
156-0659-XX	U585	8000	FF(-01)	7D(-02)	7D(-02)	7D(-02)	7D(-02)	7D(-00)	U581	160-0909-XX
156-0660-XX	U581	8800	EO(-01)	EO(-01)	EO(-01)	EO(-01)	EO(-01)	EO(-00)	U481	160-0908-XX
156-0661-XX	U487	9000	5B(-01)	5B(-01)	5B(-01)	5B(-01)	5B(-01)	5B(-00)	U481	160-0908-XX
156-0662-XX	U485	9800	1C(-01)	1C(-01)	1C(-01)	1C(-01)	1C(-01)	1C(-00)	U481	160-0908-XX
156-0663-XX	U481	A000	8A(-01)	8A(-01)	8A(-01)	8A(-01)	8A(-01)	8A(-00)	U481	160-0908-XX
156-0664-XX	U385	A800	AF(-01)	53(-02)	53(-02)	53(-02)	53(-02)	53(-00)	U581	160-0909-XX
156-0665-XX	U381	B000	B2(-01)	B2(-02)	B2(-02)	B2(-02)	B2(-02)	B2(-00)	U581	160-0909-XX
156-0666-XX	U285	B800	4D(-01)	4D(-01)	4D(-01)	4D(-01)	4D(-01)	4D(-00)	U581	160-0909-XX
156-0667-XX	U595	C000	BE(-01)	BE(-02)	BE(-02)	BE(-02)	BE(-02)	BE(-00)	U385	160-0907-XX
156-0668-XX	U591	C800	42(-01)	42(-01)	42(-02)	42(-02)	42(-01)	42(-00)	U385	160-0907-XX
156-0669-XX	U497	D000	E2(-01)	E2(-01)	E2(-01)	E2(-01)	E2(-01)	E2(-00)	U385	160-0907-XX
156-0670-XX	U495	D800	F4(-01)	F4(-01)	F4(-01)	F4(-01)	F4(-01)	F4(-00)	U385	160-0907-XX
156-0671-XX	U491	E000	E2(-01)	E2(-02)	E2(-02)	E2(-02)	E2(-02)	E2(-00)	U381	160-0906-XX
156-0672-XX	U395	E800	59(-01)	59(-01)	AD(-02)	AD(-02)	AD(-02)	AD(-00)	U381	160-0906-XX
156-0673-XX	U391	F000	10(-01)	CE(-02)	CE(-02)	CE(-03)	CE(-03)	CE(-00)	U381	160-0906-XX
156-0674-XX	U295	F800	F4(-01)	F4(-02)	F4(-02)	F4(-02)	F4(-02)	F4(-00)	U381	160-0906-XX

Checking the 4051 Backpack ROM Devices

NOTE

There are two versions of both the Option 1 Communications Backpack and the Firmware Backpack. The test procedures differ for each so each is listed separately. The standard version test is detailed first.

1. To check backpack ROM devices, remove the ROM chips from the backpack and substitute them for ROMs on the CPU board (TEKTRONIX part number 670-4008-XX). Repeat Steps 1 through 9 of the 4051 ROM Test procedure (described previously in this section). Substitute the checksums in Tables 5-4 and 5-5 for the ROM locations replaced. The numbers in parentheses are part number suffix indicators.

NOTE

DO NOT REPLACE U585 ON THE CPU BOARD FOR THIS TEST.

The numbers in parentheses are part number suffix indicators.

Table 5-4

CHECKSUMS FOR BACKPACK ROMS

Firmware/Communications (Overflow ROMs)	Version 1	Version 2
U101/U1 (156-0747-XX) U201/U11 (156-0748-XX)		EA(-01) 38(-01)

Table 5-5

COMMUNICATION BACKPACK (Communication ROMs)

Overflow ROMs	Version 1	Version 2
U101 (156-0712-XX)	29(-00)	29(-01)
U111 (156-0713-XX)	43(-00)	93(-01)
U121 (156-0714-XX)	80(-00)	C1(-01)
U131 (156-0715-XX)	EA(-01)	EA(-01)

2. The alternate Firmware Backpack (TEKTRONIX part number 670-7091-XX) and the alternate Option 1 Communications Backpack (TEKTRONIX part number 670-7069-XX) may be tested in a similar manner in the alternate CPU (TEKTRONIX part number 670-7086-XX).

To check the backpack ROM devices, remove the part(s) from the backpack and substitute for U385 on the CPU board. Repeat steps 1 through 9 of the 4051 ROM Test procedure and substitute the checksums in Tables 5-6 and 5-7.

NOTE

Each alternate ROM now has four checksums. All four checksums must be correct or the part is defective.

Table 5-6

**CHECKSUMS FOR BACKPACK ROM
(Substituted for U385)**

Firmware/Communications	Address	Version 6
U101/U111 (160-0910-XX)	C000	38(-00)
U101/U111 (160-0910-XX)	C800	EA(-00)
U101/U111 (160-0910-XX)	D000	FF(-00)
U101/U111 (160-0910-XX)	D800	F6(-00)

Table 5-7

**CHECKSUMS FOR COMMUNICATION ROM
(Substituted for U385)**

Communications	Address	Version 6
U121 (160-0911-XX)	C000	EA(-00)
U121 (160-0911-XX)	C800	29(-00)
U121 (160-0911-XX)	D000	93(-00)
U121 (160-0911-XX)	D800	C1(-00)